

**PRAGATI ENGINEERING COLLEGE: SURAMPALEM
(AUTONOMOUS)**

I B.Tech I Semester Regular Examinations, January-2024

INTRODUCTION TO PROGRAMMING

(Common to all branches)

Time: 3 hours

Max. Marks: 70

Note:

- i. Question No. 1 shall contain 10 compulsory short answer questions (2 questions from each unit) for a total of 20 marks such that each question carries 2 marks.
- ii. In each of the questions from 2 to the last question, there shall be either/or type questions of 10 marks each.

Q. No.	Questions	BTL	CO	Marks
1.	a) What is the purpose of Program Counter in Computer System?	K1	CO1	2M
	b) Define Time Complexity.	K1	CO1	2M
	c) List the loop control statements supported by C.	K1	CO2	2M
	d) Write the syntax of 'switch' statement in C.	K3	CO2	2M
	e) Define Array.	K1	CO3	2M
	f) Specify the significance of NULL character in C.	K1	CO3	2M
	g) What are Reference and Dereference operators in C?	K1	CO4	2M
	h) How to declare a Pointer to Pointer in C?	K1	CO4	2M
	i) Specify the functions to read a character from a file in C.	K1	CO5	2M
	j) Define Recursion.	K1	CO5	2M
UNIT-I				
2.	a) Discuss the types of Computer Memory.	K2	CO1	5M
	b) How the precedence and Associativity rules of operators help in executing a 'C' expression? Explain with an example.	K2	CO1	5M
OR				
3.	a) Write an algorithm to generate the Fibonacci series up to N terms.	K3	CO1	5M
	b) Explain about Implicit and Explicit Type Conversion in 'C' programming.	K2	CO1	5M
UNIT-II				
4.	a) Write the syntax of various multi-way selection statements supported by C programming.	K3	CO2	5M
	b) How 'continue' and 'break' statements affect the execution flow of 'for' loop in C? Explain with an example program.	K3	CO2	5M
OR				
5.	a) Explain about Entry-controlled and Exit-controlled loop statements in C programming with necessary flowcharts.	K3	CO2	5M

	b)	Write a C program to generate Pascal's triangle. <i>Hint:</i> The pattern for Pascal's triangle of 4 th row can be observed as given below. 1 1 1 1 2 1 1 3 3 1	K3	CO2	5M
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UNIT-III

6.	a)	How to declare, define and access multidimensional arrays in C? Explain with an example.	K3	CO3	5M
	b)	Write about standard String handling functions in C.	K3	CO3	5M

OR

7.	a)	Write a C program to find the transpose of a matrix. <i>Hint:</i> Input matrix Transpose matrix 1 4 9 1 5 5 2 6 4 2 9 6	K3	CO3	5M
	b)	How to initialize one dimensional and two dimensional arrays? Explain with examples.	K2	CO3	5M

UNIT-IV

8.	a)	Define Pointer. How it is different from normal variables? Why do Pointers have a data type?	K2	CO4	5M
	b)	Explain the concept of structures and unions with suitable examples.	K2	CO4	5M

OR

9.	a)	Explain the importance of Scale factor in Pointer arithmetic with an example program.	K2	CO4	5M
	b)	What is the use of 'typedef' keyword? Explain with programming example.	K3	CO4	5M

UNIT-V

10.	a)	Write a C program to generate Fibonacci series using recursive functions.	K3	CO5	5M
	b)	How many Storage classes are there in C programming? Explain the characteristics of each with a C program	K3	CO5	5M

OR

11.	a)	List and Explain various functions used for opening, closing and processing files in C.	K2	CO5	5M
	b)	Differentiate between Call by Value and Call by Reference.	K2	CO5	5M